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# Feature Selection

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# Feature Selection

# Introduction

## Feature Selection



- Another way to reduce the dimensions from data is selecting a subset of features

- Feature selection involves finding a subset of $k$ out of $m$ dimensions that give the most information (or appear to be irrelevant)
  - The technique **discards** the other $(m - k)$ dimensions

- The best subset contains the least number of dimensions that most contribute to the predictive ability

# Introduction

## Feature Selection



- There are $2^m$ possible subsets of $m$ variables

- Classical approaches
  - Sequential Forward Selection
  - Sequential Backward Selection

- Forward and backward algorithms belong to the family of **wrapper approaches**
  - The process of feature selection is thought to *wrap* around the learner it uses as a subroutine

# Introduction

## Feature Selection



- Advanced approaches
  - Infinite feature selection (model-agnostic) (Roffo et al., 2020)
  - SHapley Additive exPlanations (Lundberg et al. 2017) for feature importance ranking

Roffo et al. *Infinite Feature Selection: a Graph-based Feature Filtering Approach*. IEEE PAMI, 2020

Lundberg et al. *A Unified Approach to Interpreting Model Predictions.* NeurIPS, 2017

# Preliminaries

## Feature Selection



- Assume $F$ a subset of features
  - $|F| \leq m$

- Let $\mathcal{M}(F)$ be a measure of the predictive performance of a given model using $F$
  - For example, accuracy for classification and 1-error for regression
  - The higher the $\mathcal{M}(.)$, the better the predictive performance

- Best subset selection (Oracle)
  - Learn a model with all possible subsets $\binom{m}{k}$ of size $k$
  - Select the subset with highest $\mathcal{M}$

# Sequential Forward Selection

## Feature Selection



- General idea
  - Start with no features: $F = \emptyset$
  - At each step, for every candidate feature $x^i$, train the model on the training set and compute $\mathcal{M}(F \cup x^i)$ on the **validation set**
  - Select the feature $x^i$ that yields the best $\mathcal{M}$

$F \leftarrow \emptyset$

While the stopping criteria are not satisfied

$$i^* = \operatorname*{argmax}_{x^i \notin F} \mathcal{M}\left(F \cup \{x^i\}\right)$$

$$F \leftarrow F \cup \{x^{i^*}\}$$

$\mathcal{M}(x^1) = 0.3$ $\mathcal{M}(x^2) = 0.5$ $\boldsymbol{\mathcal{M}(x^3) = 0.65}$ $\mathcal{M}(x^4) = 0.4$

$\mathcal{M}(\{x^3, x^1\}) = 0.6$ $\boldsymbol{\mathcal{M}(x^3, x^2\}) = 0.75}$ $\mathcal{M}(\{x^3, x^4\}) = 0.7$

$\mathcal{M}(\{x^3, x^2, x^1\}) = 0.77$ $\boldsymbol{\mathcal{M}(\{x^3, x^2, x^4\}) = 0.78}$

$\mathcal{M}(\{x^3, x^2, x^4, x^1\}) = 0.8$

# Sequential Backward Selection

## Feature Selection



- General idea
  - Start with all features: $F = \{x^1, x^2 \dots x^m\}$
  - At each step, for every feature $x_i \in F$, train the model on the training set after removing $x^i$ and compute $\mathcal{M}(F \setminus x^i)$ on the **validation set**
  - Select the feature $x^i$ whose removal yields the best $\mathcal{M}$

$F \leftarrow \{x^1, x^2 \dots x^m\}$

While the stopping criteria are not satisfied

$$i^* = \underset{x^i \in F}{\operatorname{argmax}} \mathcal{M}\left(F \setminus \{x^i\}\right)$$

$$F \leftarrow F \setminus \{x^{i^*}\}$$

# Stopping criteria

## Feature Selection

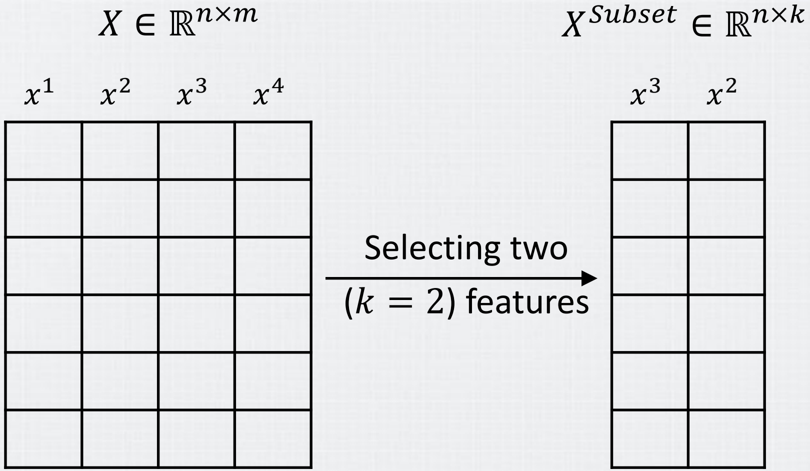
- Number of specified features
- If adding or removing any feature does not change $\mathcal{M}$ by more than a threshold
- If the improvement in $\mathcal{M}$ is below a pre-defined threshold (task-specific)

# Advanced Approaches

## Feature Selection

- Overview of model-agnostic methods
    - Sort the features by importance (score)
    - Select $k$ features according to the score

| Feature | $x^1$ | $x^2$ | $x^3$ | $x^4$ |
| --- | --- | --- | --- | --- |
| Score (Importance) | 0.15 | 0.25 | 0.37 | 0.23 |




Roffo et al. *Infinite Feature Selection: a Graph-based Feature Filtering Approach*. IEEE PAMI, 2020

# Difference from Dimensionality Reduction

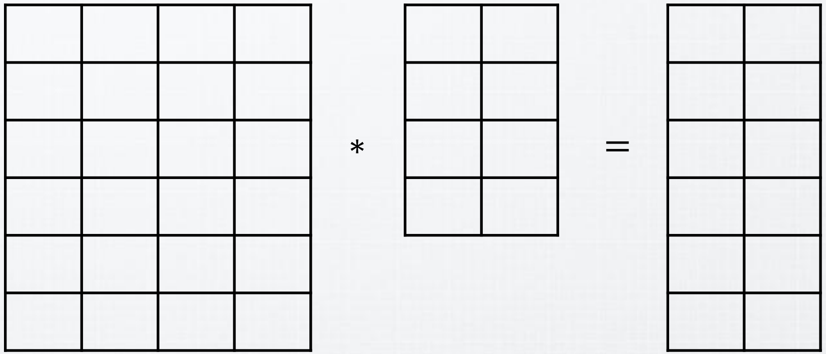
## Feature Selection



- Assume $c$ and $|F|$ as 2

### Dimensionality Reduction

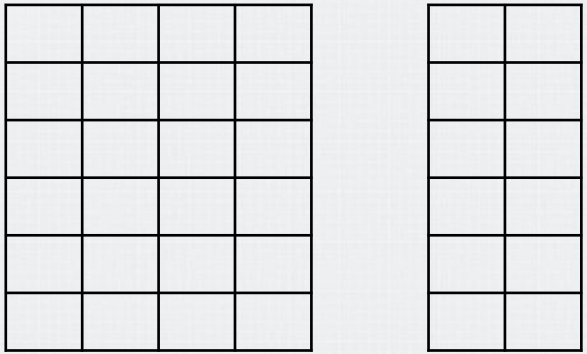
$X \in \mathbb{R}^{n \times m}$ $\quad$ $W \in \mathbb{R}^{m \times c}$ $\quad$ $X^{Latent} \in \mathbb{R}^{n \times c}$




We must always **project** (matrix multiplication) both the training and test data

### Feature Selection

$X \in \mathbb{R}^{n \times m}$ $\quad$ $X^{Subset} \in \mathbb{R}^{n \times |F|}$



Just indexes the kept features

# Bibliography

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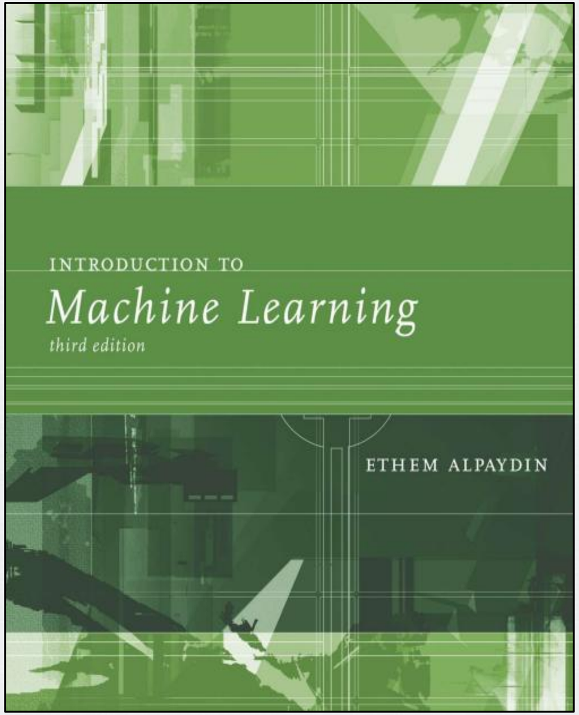
- INTRODUCTION TO Machine Learning
  - Chapter 6
    - 6.2 Subset Selection



# Bibliography

- An Introduction to Statistical Learning
    - Chapter 6
        - 6.1 Subset Selection